From Visual Affinity to Learning Credit in Multimodal Reinforcement Learning

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Abstract

Reinforcement learning with verifiable rewards (RLVR) improves multimodal reasoning, but uniformly assigning a single response-level advantage to all tokens overlooks the distinct roles of visual observations and subsequent reasoning. Visual affinity alone is insufficient: predictive states may align with both image and prompt-context representations, and upweighting valid visual observations can amplify their negative-advantage updates when later reasoning fails. To address this issue, we propose ***Outcome-Conditioned Visual Credit (OVC)****, a plug-in framework that separates visual-association measurement from token-level credit assignment and is adaptable to policy-optimization methods with uniform response-level advantage allocation. Relative Visual Encoding constructs a context-relative visual coordinate by contrasting each predictive state's affinity to image and prompt-context representations. Outcome-conditioned Evidence Calibration combines this coordinate with uncertainty-scaled predictive features and calibrates their contributions to outcome-discriminative evidence through mean contrasts between positive- and negative-advantage responses, scaled by regularized within-side variation. Visual association thus informs credit through observed outcome differences rather than directly determining update priority. Normalized Credit Allocation conditions the calibrated evidence on the response's advantage sign to produce positive token weights with unit global mean. These detached weights preserve advantage signs while modulating individual token-gradient contributions. OVC reuses existing rollouts without auxiliary reward models, process annotations, or inference-time modules. Instantiated through GSPO-token, OVC retains sequence-level clipping and consistently improves six-benchmark mean accuracy over GSPO across three models, with gains of 1.86–2.39 percentage points.*

1. Introduction

Reinforcement learning with verifiable rewards (RLVR) [9, 19, 37, 42] has become a key approach to improving reasoning in large vision language models (LVLMs) [1, 22, 31]. It provides supervision from verifiable outcomes without requiring dense annotations of intermediate steps. Recent efforts have advanced multimodal reasoning through improved training data [2, 8, 11–13, 16, 34], policy optimization [6, 20, 23, 26, 32, 33, 36], and perceptual supervision [27, 29]. Yet the granularity of outcome feedback remains coarser than the computations it guides. A single response may involve visual observations, symbolic transformations, and intermediate inferences, while its reward evaluates only the complete solution. This mismatch has motivated finer-grained credit assignment that uses visual signals to differentiate token updates.

Existing approaches [5, 7, 9, 30] derive token-level signals from visual dependence, representational similarity, predictive uncertainty, or region alignment. Although these signals capture different aspects of visual involvement, their conversion into update coefficients still requires deciding how the measured association should affect learning. With DelTA's predictive-state representation [40], a state may align with both image and prompt representations, so absolute image affinity alone does not distinguish these associations. Moreover, a valid visual observation may occur in a negative-advantage response when later reasoning fails. Upweighting it would strengthen its negative update even if the error occurs elsewhere in the response. The central issue is therefore **not merely how to measure visual association, but how to interpret it under outcome feedback.**

Figure 1 illustrates this distinction. In (a), the highest image-affinity group does not attain the highest observed mean accuracy, suggesting a possible mismatch between affinity ranking and response quality. In (b), responses approximately matched on image affinity exhibit different outcome statistics across context-affinity levels. These observations motivate considering contextual association alongside image affinity, rather than treating the latter alone as sufficient for credit allocation. Together with the advantage-sign dependence discussed above, they motivate a two-stage principle: measuring visual relevance relative to context and calibrating it against outcome differences.

To instantiate this principle, we propose **Outcome-Conditioned Visual Credit (OVC)**. Its first component, **Rel-**

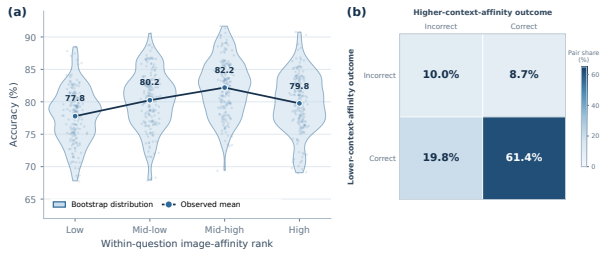


Figure 1. **Affinity and response correctness.** We analyze eight valid responses per randomly sampled question from a frozen Qwen3.5-9B trained with GSPO on Math77K, without correctness filtering. Affinity is $s_i^u = |\mathcal{R}_i|^{-1} \sum_{t \in \mathcal{R}_i} \cos(h_{it}, g_i^u)$, where h_{it} predicts token t, g_i^u is the mean-pooled image ($u = \text{img}$) or non-image prompt ($u = \text{ctx}$) prototype from the same intermediate layer, and $\mathcal{R}_i$ indexes reasoning tokens excluding special tokens and the final answer. No coordinate rescaling or image–context subtraction is applied. **(a)** Mean accuracy across four within-question image-affinity rank groups (two responses each). Each faint point is a group mean from a question-level bootstrap resample, with shared indices across groups. Violins summarize these distributions; dark markers show original-sample means. **(b)** Joint correctness of within-question pairs approximately matched on image affinity. Rows and columns represent lower- and higher-context-affinity response outcomes, respectively; cells report percentages of all matched pairs.

ative Visual Encoding (RVE), contrasts a predictive state's cosine affinity to image and prompt-background representations under a shared diagonal scaling. The resulting coordinate captures relative visual association without directly prescribing an update magnitude. **Outcome-conditioned Evidence Calibration (OEC)** then combines this coordinate with predictive features and determines their evidential contributions from contrasts between positive- and negative-advantage responses. A regularized estimate of within-side diagonal variation calibrates the relative contribution of each coordinate. Consequently, with the statistics held fixed, the visual contribution depends on the outcome contrast and the variance scale, not on affinity alone, and equals zero when the visual reference means coincide.

This calibrated evidence is converted into policy updates through **Normalized Credit Allocation (NCA)**. NCA produces positive, globally normalized token coefficients that modulate the original sequence advantage while preserving its sign. Applied through GSPO-token [42], these detached coefficients retain sequence-level importance ratios and clipping while adjusting the contribution of individual token-gradient terms. OVC introduces no auxiliary reward model, additional process or region supervision, or inference-time module. To clarify this reweighting, we characterize the resulting gradient and separate response-level weighting from within-response allocation, enabling controlled evaluation of token–coefficient correspondence.

Our contributions are threefold.

- We introduce a context-relative visual coordinate that contrasts image affinity with prompt-background affinity, providing an explicit visual measurement for credit estimation.
- We develop an outcome-conditioned calibration mechanism that combines visual and predictive features through advantage-side contrasts and regularized within-side variation, without assigning visual affinity a fixed positive update preference.
- We characterize how calibrated evidence enters policy updates and distinguish response-level reweighting from allocation within a response, establishing an analytical basis for controlled evaluation of token credit assignment.

2. Related Work

Policy Optimization and Perceptual Supervision for Multimodal Reasoning. RLVR improves reasoning through outcome verification. GRPO [19] employs group-relative advantage estimation, while GSPO [42] introduces sequence-level importance ratios for policy optimization. In multimodal settings, MM-Eureka [16] investigates large-scale training with rule-based rewards, and VL-Rethinker [24] improves training through selective sample replay and forced reflection. Another line of work refines supervision. PAPO [29] constructs an implicit perceptual objective through image perturbations, whereas VisualPRM [27] trains a separate process reward model to evaluate intermediate steps and support candidate solution selection. These studies primarily advance optimization mechanisms, training procedures, or supervision sources. Complementing these efforts, our work investigates how visual representations can inform fine-grained credit estimation under outcome feedback, rather than redesigning sampling strategies or introducing an independent process evaluator.

Visually Guided Fine-Grained Credit Assignment. VPPO [7] and PGPO [35] exploit changes in predictive distributions across visual conditions to filter token updates or redistribute advantages. PRPO [10] further combines strong and weak perturbations to distinguish visual dependence from prediction instability. VGPO [30] derives attention-related scores from visual prototypes and combines late-stage compensation with advantage reweighting within and across trajectories. `KAWHI` [5] connects structured visual evidence to RLVR through geometry-guided key-region selection, query-key alignment in vision-critical heads, and paragraph-level credit redistribution. These methods explicitly connect visual signals to policy updates. However, visual sensitivity, representational affinity, and region alignment do not, by themselves, determine a token's contribution to outcome improvement. Our approach instead treats visual association as a feature requiring calibration. We construct

a context-relative visual coordinate and determine its evidential contribution through statistical contrasts between positive- and negative-advantage responses and within-side variance, thereby separating the measurement of visual relevance from the allocation of learning credit.

3. Method

Visual relevance describes an association with the image, whereas learning credit determines the strength of a policy update. We treat the former as a measurement to be calibrated against outcome feedback, rather than as a quantity to maximize. **Outcome-Conditioned Visual Credit (OVC)** implements this principle through Relative Visual Encoding (RVE), Outcome-conditioned Evidence Calibration (OEC), and Normalized Credit Allocation (NCA). We first characterize the calibration rule, then specify its estimator and policy objective.

Setting and notation. For each multimodal input $x = (I, q)$, a behavior policy π_{old} samples G responses. Response $y_i = (y_{it})_{t=1}^{T_i}$ receives a verifiable reward R_i and a group-relative advantage [19]

$$A_i = \frac{R_i - \overline{R}_{g(i)}}{s_{g(i)} + \epsilon_A}, \qquad c_i = \text{sign}(A_i), \tag{1}$$

where $g(i)$ contains responses to the same input, $\overline{R}_{g(i)}$ and $s_{g(i)}$ are the corresponding reward mean and standard deviation, and $\epsilon_A > 0$. The sign c_i indicates relative reward, not token correctness. An actor mini-batch contains B nonempty responses and $N = \sum_i T_i$ valid tokens. Advantages are computed within each rollout group, whereas the reference statistics and coefficient normalization are aggregated over the actor mini-batch. All credit statistics are detached and fixed during its optimizer step. Sums below exclude padding.

3.1. A Calibration Principle for Visual Credit

Let $z_{it} = [v_{it}; \rho_{it}]$ combine a predictive feature v_{it} with a scalar visual measurement ρ_{it}. For $c \in \{-1, +1\}$, let $\mathcal{T}_c = \{(i,t) \mid c_i = c,\ 1 \le t \le T_i\}$ contain the valid response tokens on the corresponding advantage side. We initialize the reference means using advantage-magnitude weights

$$\mu_c^{(0)} = \frac{1}{M_c} \sum_{(i,t)\in\mathcal{T}_c} |A_i| z_{it}, \qquad M_c = \sum_{(i,t)\in\mathcal{T}_c} |A_i|, \quad M_c > 0. \tag{2}$$

These statistics are aggregated across the actor mini-batch, rather than estimated separately for each input. The reference-based calibration below assumes $M_+ > 0$ and $M_- > 0$. The two sides are defined by the sign of the sequence advantage, rather than by token-level correctness labels.

We write μ_+, μ_- generically for the reference means before or after refinement. Their refinement and the construction of a positive diagonal matrix D encoding regularized within-side variation are specified in Section 3.2. We calibrate a token feature against both advantage sides by comparing its squared distances to the two references under the shared metric D^{-1}:

$$e(z) = \|z - \mu_-\|_{D^{-1}}^2 - \|z - \mu_+\|_{D^{-1}}^2, \tag{3}$$

where $\|a\|_{D^{-1}}^2 = a^\top D^{-1} a$. Positive evidence indicates that z is closer to the positive-advantage reference, while negative evidence indicates the opposite.

For fixed reference statistics, define the variance-calibrated direction

$$b^* = D^{-1}\Delta\mu, \qquad \Delta\mu = \mu_+ - \mu_-.$$

Direct expansion of the distance contrast in Eq. (3) gives

$$e(z) = 2(b^*)^\top (z - \overline{\mu}), \qquad \overline{\mu} = \frac{\mu_+ + \mu_-}{2}. \tag{4}$$

Each coordinate therefore contributes through its advantage-side mean contrast relative to its regularized within-side scale, rather than through its magnitude alone. For $\Delta\mu \neq 0$, Appendix C further shows that b^* maximizes the squared reference separation normalized by a regularized diagonal variation surrogate, up to nonzero rescaling. This provides a fixed-statistic characterization of the calibration rule, not an auxiliary objective optimized by the actor.

When does visual relevance influence credit? With $D = \text{diag}(D_h, d_\rho)$, Eq. (3) decomposes as

$$e(z) = e_h(v) + 2 \underbrace{\frac{\Delta\mu_\rho}{d_\rho}}_{\text{outcome-dependent coefficient}} (\rho - \overline{\mu}_\rho). \tag{5}$$

For fixed statistics, the direct visual term vanishes when $\Delta\mu_\rho = 0$ and reverses its dependence on ρ when the contrast changes sign. We use the signed evidence $c_i e(z_{it})$ to measure agreement with the response's update direction. Thus, larger relative visual affinity does not receive an unconditional positive credit bonus. Appendix C.3 derives the conditional sensitivities of the evidence and raw coefficients and distinguishes them from causal attribution. This motivates separating visual-relevance measurement from credit assignment.

3.2. Estimating Relative Visual Evidence

Relative Visual Encoding. Let $h_{it} \in \mathbb{R}^H$ be the state at a fixed intermediate layer aligned with the prediction of y_{it}. Let H_{is} denote states at the same layer over the full

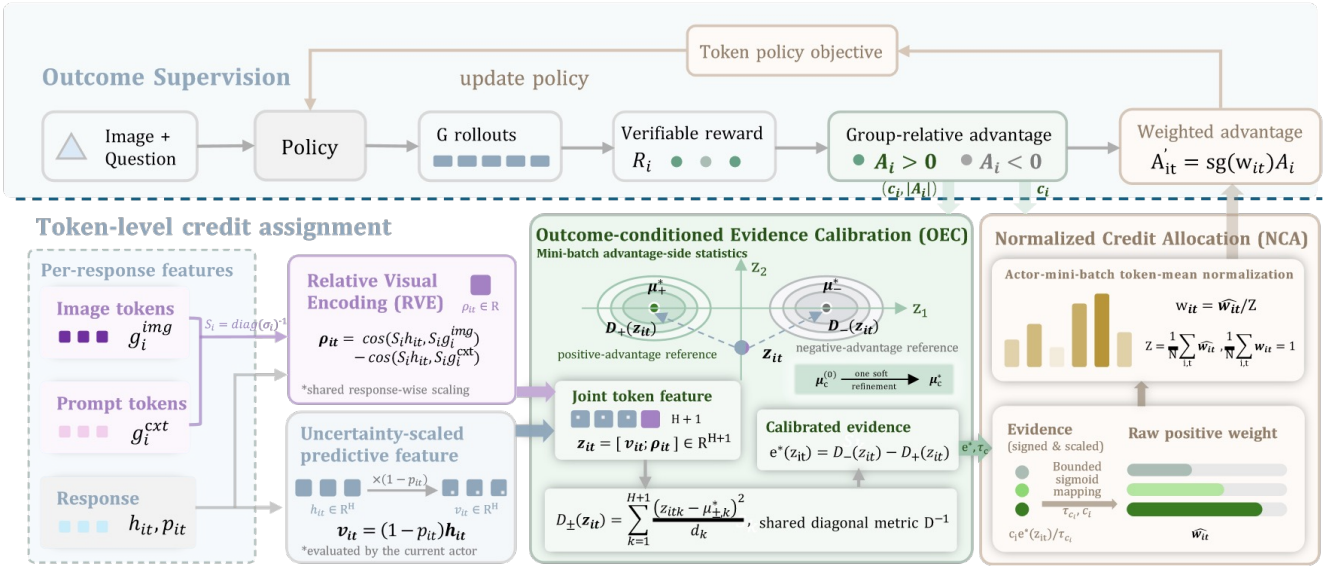


Figure 2. **Overview of Outcome-Conditioned Visual Credit (OVC).** An image–question pair produces a group of responses whose rewards yield group-relative advantages. For each response token, OVC combines relative image–context affinity with a prediction-aware representation, calibrates this evidence across advantage groups, and assigns normalized token weights to the policy update.

input sequence. Image positions $\mathcal{V}_i$ and non-image prompt positions $\mathcal{C}_i$ define

$$g_i^{\text{img}} = \frac{1}{|\mathcal{V}_i|}\sum_{s\in\mathcal{V}_i} H_{is}, \qquad g_i^{\text{ctx}} = \frac{1}{|\mathcal{C}_i|}\sum_{s\in\mathcal{C}_i} H_{is}. \quad (6)$$

The context prototype is obtained from the original multimodal input, not from a separate text-only forward. Let $S_i = \text{diag}(\sigma_i)^{-1}$, where σ_i contains floored response-wise population standard deviations of h_{it}. Applying the same scaling to the predictive state and both prototypes, RVE computes

$$\rho_{it} = \cos(S_i h_{it}, S_i g_i^{\text{img}}) - \cos(S_i h_{it}, S_i g_i^{\text{ctx}}). \quad (7)$$

This measures image affinity relative to prompt context. The mean used to estimate σ_i is not subtracted in the cosine calculation. Appendix B gives the estimator and its metric interpretation.

We retain sampled-token prediction information with an uncertainty-scaled feature

$$v_{it} = (1 - p_{it})h_{it}, \qquad z_{it} = [v_{it}; \rho_{it}], \quad (8)$$

where $p_{it} = \pi_\theta(y_{it} \mid x_i, y_{i,<t})$ is evaluated by the current actor. Only v_{it} is multiplied by $1 - p_{it}$. Using a state from the selected intermediate layer makes v_{it} a gradient-inspired predictive feature rather than an exact parameter gradient (Appendix B.2).

Outcome-conditioned Evidence Calibration. Given $\mu_c^{(0)}$ and M_c from Eq. (2), OEC estimates advantage-weighted within-side variation over the actor mini-batch:

$$q_k = \sum_{c\in\{-1,+1\}} \frac{1}{M_c} \sum_{(i,t)\in\mathcal{T}_c} |A_i| \left(z_{itk} - \mu_{c,k}^{(0)}\right)^2. \quad (9)$$

For all $H+1$ coordinates, we construct the diagonal scale matrix $D = \text{diag}(d)$ using shrinkage and a variance floor:

$$\bar{q} = \frac{1}{H}\sum_{k=1}^{H} q_k, \qquad d_k = \max\left\{(1-\eta)q_k + \eta\bar{q} + \epsilon_D,\ r_{\min}\bar{q} + \epsilon_D\right\}. \quad (10)$$

Here $0 \le \eta \le 1$, $r_{\min} \ge 0$, and $\epsilon_D > 0$, so $d_k \ge \epsilon_D$ and $D \succ 0$. The visual coordinate is excluded from the shrinkage target $\bar{q}$ but receives the same regularization. This is a diagonal scale estimate, not a full covariance matrix.

We perform one soft reference refinement. Let $e^{(0)}$ use the initial references in Eq. (4), and define τ_c as the unweighted population standard deviation of $ce^{(0)}(z_{it})$ over $\mathcal{T}_c$, with a strictly positive floor. We then compute

$$u_{it}^{(0)} = \text{sigmoid}\left(\frac{c_i e^{(0)}(z_{it})}{\tau_{c_i}}\right), \qquad \mu_c^* = \frac{\sum_{\mathcal{T}_c} |A_i| u_{it}^{(0)} z_{it}}{\sum_{\mathcal{T}_c} |A_i| u_{it}^{(0)}}. \quad (11)$$

The final score e^* uses μ_+^*, μ_-^* and the same D; the final coefficient mapping also retains the pre-refinement τ_c. This makes the visual coefficient in Eq. (5) outcome-dependent after refinement as well. Appendix D specifies the ordering and derives the soft assignment and reference update.

3.3. Credit Allocation and Policy Optimization

Normalized Credit Allocation. For $0 < \lambda_{\min} \le \lambda_{\max}$, NCA maps signed evidence to positive raw coefficients and

applies token-mean normalization over the actor mini-batch:

$$\widehat{w}_{it} = \lambda_{\min} + (\lambda_{\max} - \lambda_{\min})\,\mathrm{sigmoid}\left(\frac{c_i e^*(z_{it})}{\tau_{c_i}}\right),$$
$$Z = \frac{1}{N}\sum_{i,t}\widehat{w}_{it}, \qquad w_{it} = \widehat{w}_{it}/Z, \qquad A'_{it} = \mathrm{sg}(w_{it})A_i. \tag{12}$$

For $A_i = 0$, we instead set $\widehat{w}_{it} = \lambda_{\min}$ and include it in Z. Positive coefficients preserve advantage signs. Appendix E.1 gives their final bounds. Normalization is global rather than per response, so the estimator can adjust both relative response strength and allocation within a response. It fixes the global mean coefficient, but does not constrain the norm of the resulting policy gradient.

We apply these coefficients through GSPO-token. With $L_{it}(\theta) = \log \pi_\theta(y_{it} \mid x_i, y_{i,<t})$, define

$$s_i(\theta) = \exp\left[\frac{1}{T_i}\sum_t \big(L_{it}(\theta) - L_{it}(\theta_{\mathrm{old}})\big)\right],$$
$$\widetilde{s}_{it}(\theta) = \mathrm{sg}(s_i(\theta))\exp\big(L_{it}(\theta) - \mathrm{sg}(L_{it}(\theta))\big). \tag{13}$$

Here sg stops gradients. The surrogate to maximize is

$$\mathcal{J}_{\mathrm{OVC}}(\theta) = \frac{1}{B}\sum_i \frac{1}{T_i}\sum_t \min\left\{\widetilde{s}_{it}A'_{it}, \mathrm{clip}(\widetilde{s}_{it}, 1-\epsilon_l, 1+\epsilon_h)A'_{it}\right\}. \tag{14}$$

Although $\widetilde{s}_{it} = s_i$ numerically, its derivative remains token-specific. Appendices E.2 and E.3 derive this gradient and separate response-level weighting from token allocation. No gradients pass through the credit estimator. The policy loss can still update the entire actor, including its visual encoder. OVC reuses the sampled responses and adds no learnable head or inference-time module. Algorithm 1 and Appendix F describe the actual multi-pass training computation.

4. Experiment

4.1. Experimental Setup

Models and training data. We evaluate Qwen3-VL-4B-Instruct, Qwen3-VL-8B-Instruct [1], and Qwen3.5-9B [22]. The Qwen3-VL models are trained on ViRL39K [24], while Qwen3.5-9B uses Math77K, the 77,135-example mathematical subset of DeepVision-103K [21]. All methods share the same instruction-tuned initialization at each scale.

Benchmarks. We report MMMU [38], MathVerse [41], MathVision [25], We-Math [17], MathVista [14], and MMK12-val [16]. The first five use lmms-eval [39] with matched decoding and scoring within each model size. MMK12-val follows the fixed training-time validation protocol. Full settings are given in Appendix A.

4.2. EXPERIMENTAL RESULTS

4.2.1. Main Results

We compare with the instruction-tuned Base model, GSPO [42], DelTA [40], VPPO [7], and AT-RL [9]. OVC applies its detached weights through GSPO-token; uniform weights recover the GSPO gradient (Appendix E.2).

Consistent gains across model scales. Table 1 shows that OVC improves the six-task average over GSPO by +2.04, +2.39, and +1.86 points on 4B, 8B, and 9B, respectively, and exceeds Base by +4.80, +4.75, and +3.32 points. OVC also achieves the highest average among all compared RL methods on both Qwen3-VL scales, with the gain extending to Qwen3.5-9B trained on the larger-scale Math77K corpus.

Performance across multimodal benchmarks. The largest gains over GSPO occur on MMK12-val, MathVision, and MMMU: +7.44/+4.21 points on MMK12-val for 4B/8B, +5.59 on MathVision for 9B, and +2.16/+5.00/+3.55 on MMMU. OVC also reverses the Base-to-GSPO drop on MMMU for 4B and 8B, reaching 46.83 and 49.56 versus Base scores of 46.56 and 48.78. On 9B, the only regression relative to GSPO is MathVerse (−0.94 points).

Comparison with non-visual discriminative credit. OVC exceeds DelTA's six-task average by +0.95 points on 4B and +1.06 on 8B. It leads DelTA on five benchmarks and ties on MMK12-val at 4B. At 8B, its largest gains are MathVision (+5.92) and MMK12-val (+1.88), while DelTA remains stronger on We-Math and nearly tied on MathVista.

Gains from strong initializations. The improvements should be interpreted in the context of strong, already post-trained models. Prior exposure to related visual-mathematical problems may reduce the marginal gains from additional RL, although sample-level overlap with the original training corpora remains unverified. Under matched initializations, RL data, and rollout configurations, OVC's gains are consistent with more effective use of existing representations and outcome feedback through finer-grained credit allocation. These results suggest that improved credit assignment remains beneficial even when substantial task-relevant competence is already present.

4.2.2. Ablation studies

Table 2 examines OVC on Qwen3-VL-8B across relative visual encoding, within-side variance calibration, and token-specific weight allocation. Full OVC achieves a four-task mean of 63.85, exceeding the uniform-credit GSPO reference (62.57) by 1.28 percentage points. Configuration definitions are provided in Appendix A.2.

Table 1. **Main results on six multimodal benchmarks.** Scores are percentages, averaged over four independent runs. Within each model group, **bold** and underlining indicate the top two distinct scores, respectively. Avg. denotes the arithmetic mean across six benchmarks. Red superscripts indicate gains over the corresponding Base in percentage points, calculated from the displayed averages. The Qwen3.5-9B group uses larger-scale training data. OOM denotes an out-of-memory error.

Method	MMMU	MathVision	We-Math	MathVista	MMK12-val	MathVerse	Avg.
Qwen3-VL-4B							
Base	46.56	21.71	75.52	73.83	54.73	63.12	55.91
GSPO [42]	44.67	24.58	76.52	**74.83**	67.14	64.29	58.67$^{+2.76}$
DelTA [40]	44.33	25.00	76.00	74.00	**74.58**	64.67	59.76$^{+3.85}$
VPPO [7]	45.21	23.12	76.43	74.19	69.23	64.23	58.74$^{+2.83}$
AT-RL [9]	43.52	22.78	76.33	74.22	70.22	63.45	58.42$^{+2.51}$
GSPO+OVC (Ours)	**46.83**	**26.64**	**77.05**	74.37	**74.58**	**64.80**	**60.71**$^{+4.80}$
Qwen3-VL-8B							
Base	48.78	24.67	80.23	74.40	64.46	62.84	59.23
GSPO [42]	44.56	27.30	83.33	76.10	74.73	63.53	61.59$^{+2.36}$
DelTA [40]	49.21	24.01	**85.40**	**77.03**	77.06	64.82	62.92$^{+3.69}$
VPPO [7]	49.43	27.69	84.36	76.21	75.29	64.09	62.85$^{+3.62}$
AT-RL [9]	OOM	OOM	OOM	OOM	OOM	OOM	OOM
GSPO+OVC (Ours)	**49.56**	**29.93**	83.45	76.97	**78.94**	**65.05**	**63.98**$^{+4.75}$
Qwen3.5-9B *(Larger-scale training data: Math77K)*							
Base	45.67	36.18	87.24	79.20	79.09	77.36	67.46
GSPO [42]	46.78	34.87	87.18	79.20	87.74	**77.74**	68.92$^{+1.46}$
GSPO+OVC (Ours)	**50.33**	**40.46**	**88.51**	**80.30**	**88.28**	76.80	**70.78**$^{+3.32}$

Table 2. **Ablation study of OVC on Qwen3-VL-8B.** Base and GSPO are reference results from Table 1. Scores are percentages; Avg. is the four-benchmark mean. Δ is the change from full OVC in percentage points, calculated from the displayed averages. The highest score in each benchmark and Avg. is bold. See Appendix A.2 for configuration definitions.

Configuration	**MathVision**	**MathVista**	**MathVerse**	**We-Math**	**Avg.**	**Δ**
Base	24.67	74.40	62.84	80.23	60.54	−3.31
GSPO (uniform credit)	27.30	76.10	63.53	83.33	62.57	−1.28
OVC (full)	**29.93**	**76.97**	**65.05**	**83.45**	**63.85**	—
(a) Relative visual encoding						
No visual coordinate	28.21	74.97	63.15	81.58	61.98	−1.87
Absolute image affinity	28.03	75.63	63.47	81.79	62.23	−1.62
Separate affinity coordinates	28.89	75.59	63.83	82.02	62.58	−1.27
No response-wise scaling ($S_i = I$)	28.89	74.99	63.41	82.11	62.35	−1.50
(b) Within-side variance calibration						
Identity metric ($D = I$)	28.12	75.34	63.35	81.58	62.10	−1.75
Total-variance calibration	28.09	75.91	63.45	81.64	62.27	−1.58
No visual coordinate + $D = I$	28.64	75.23	63.32	81.98	62.29	−1.56
(c) Token-specific weight allocation						
Response-mean weights	28.06	75.39	63.85	82.17	62.37	−1.48
Within-response permutation	28.22	75.14	63.53	82.27	62.29	−1.56
Affinity-ranked weights	28.44	75.14	63.39	82.11	62.27	−1.58

Relative visual encoding. Removing the visual coordinate, using absolute image affinity, or using separate image/context coordinates lowers the four-task mean by 1.87, 1.62, and 1.27 points, respectively. Separate coordinates retain both affinities yet underperform their difference, favoring an explicit image–context contrast over simply including both measurements. Removing response-wise scaling ($S_i = I$) costs 1.50 points, indicating that the pre-cosine

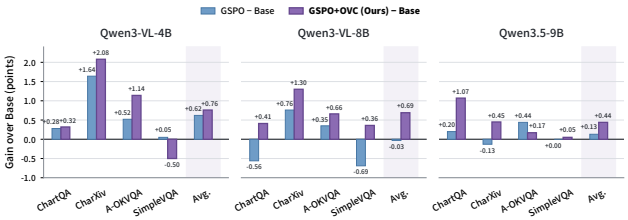


Figure 3. **Zero-shot transfer gains over Base.** Bars show score changes relative to the corresponding instruction-tuned Base model. Avg. is the mean over four VQA benchmarks.

measurement geometry matters even when subsequent evidence calibration is retained.

Within-side variance calibration. The identity metric and total-variance calibration reduce the mean by 1.75 and 1.58 points, respectively. Setting $D = I$ retains the advantage-side reference contrast but removes variance scaling. Adding the relative visual coordinate raises the mean from 61.98 to 63.85 with within-side calibration, but lowers it from 62.29 to 62.10 under the identity metric. Within-side calibration alone likewise lowers the no-visual setting from 62.29 to 61.98. Neither addition alone improves the no-visual identity reference, whereas their combination gains 1.56 points, consistent with a positive interaction between visual encoding and calibration.

Token-specific weight allocation. Response-mean, within-response permutation, and affinity-ranked weights reduce the mean by 1.48, 1.56, and 1.58 points, respectively. Response averaging preserves each response's mean coefficient; permutation additionally preserves its complete weight multiset while disrupting token–weight correspondence. Their deficits support token-specific allocation beyond response-level reweighting or nonuniform coefficients alone. Affinity-ranked weights also underperform, favoring outcome-conditioned allocation over an unconditional preference for higher visual affinity. Most ablations also fall below GSPO, indicating that the gains are not reproduced by the tested simplified or alternative configurations.

4.2.3. Zero-shot VQA generalization

We further evaluate zero-shot transfer on ChartQA [15], CharXiv [28], A-OKVQA [18], and SimpleVQA [3]. Figure 3 shows per-task and mean gains over the corresponding Base models. OVC exceeds GSPO in the four-task mean by 0.14, 0.72, and 0.31 points on 4B, 8B, and 9B, respectively, while its four-task mean remains above Base at all three scales. The largest mean separation is on 8B, where GSPO is -0.03 relative to Base and OVC is +0.69. These results suggest that the benefits of OVC extend beyond the visual-mathematical benchmarks in the main evaluation. Absolute

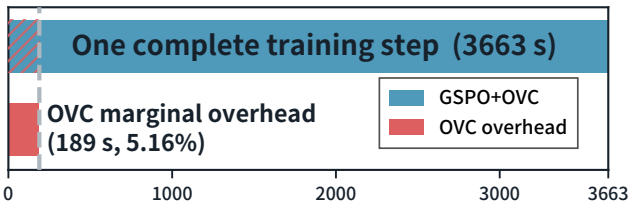


Figure 4. Training-time cost breakdown of OVC.

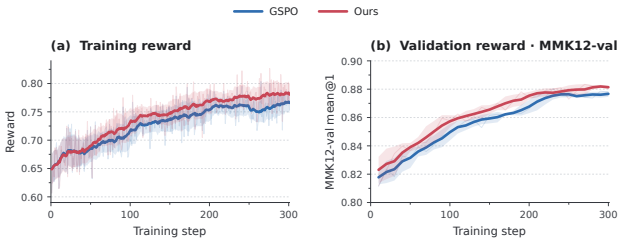


Figure 5. Training dynamics on Qwen3.5-9B.

per-task scores are reported in Appendix A.5.

4.3. Efficiency

Figure 4 shows the training-time cost breakdown. The OVC-specific credit-estimation passes account for 189 s, or 5.16% of the 3663 s total step time. OVC adds no learnable parameters or inference-time computation; its additional training cost comes from four no-gradient statistics passes. Implementation details are given in Appendix F.

4.4. Visualizations

Training dynamics. Figure 5 compares OVC and GSPO during Qwen3.5-9B training on Math77K. OVC generally achieves higher training reward and MMK12-val performance over the training trajectory. The consistent advantage in both optimization reward and validation performance provides further evidence for the effectiveness of OVC beyond the final-checkpoint comparison.

Distribution of token credit weights. Figure 6 provides a qualitative visualization of the token-credit patterns produced by OVC and DelTA. OVC correctly converts the bearing to the 65° interior angle and obtains $d = 11.8$ km. Its decisive reasoning span receives relatively high credit. In contrast, DelTA uses 25° as the triangle angle and obtains 25.3 km, while assigning relatively low credit to the first erroneous reasoning span. This example illustrates how OVC can better align token-level credit with visually grounded reasoning steps.

5. Conclusion

We introduced OVC to address a central distinction in multimodal RLVR: visual association does not, by itself, deter-

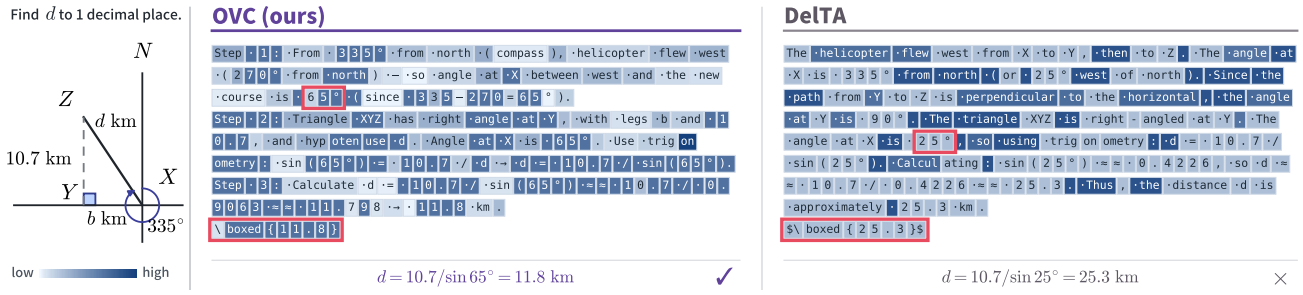


Figure 6. **Qualitative visualization of within-response token-credit patterns.** OVC correctly converts the bearing to the 65° triangle angle and obtains $d = 11.8$ km, whereas DelTA first errs by using 25° as the triangle angle and obtains 25.3 km. Colors indicate the final normalized token-credit weights w_{it}, recomputed offline from the corresponding checkpoints. Red boxes mark the key angle conversions and final answers. The OVC and DelTA responses have positive and negative group-relative advantages, respectively.

mine learning credit under response-level feedback. OVC measures visual affinity relative to context, combines it with predictive features, and calibrates their contributions through advantage-side contrasts and regularized within-side variation. The resulting evidence modulates token-specific gradient contributions through positive, normalized weights, without changing the original advantage signs or sequence-level clipping. Our analysis further distinguishes response-level reweighting from within-response allocation, clarifying how calibrated coefficients affect individual token gradients. Experiments across three models and six benchmarks show improved average accuracy over GSPO without additional process supervision or inference-time modules. Overall, OVC connects visual representations with fine-grained policy optimization by treating visual association as outcome-conditioned evidence rather than direct update priority.

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From Visual Affinity to Learning Credit in Multimodal Reinforcement Learning

Supplementary Material

A. Evaluation protocol

A.1. Benchmarks and evaluation settings

We use the official MMK12 test split as a fixed validation set and report the final scheduled validation score, and evaluate MMMU, MathVerse, MathVision, We-Math, and MathVista after training using lmms-eval. All compared checkpoints within each model size use the same benchmark splits and evaluation procedures. Table 3 summarizes the protocol.

MMMU [38]. We evaluate the 900-example validation split. Following the public Qwen protocol, all images are placed before the textual query. The prompt contains the available hint, question, and answer choices when present, and asks the model to select the correct option for multiple-choice items. Responses are scored by the benchmark's deterministic accuracy evaluator.

MathVerse [41]. We evaluate the 3,940-example testmini split with the benchmark's zero-shot query that does not add an explicit chain-of-thought instruction or few-shot demonstrations. Benchmark-defined termination strings are preserved. Responses are judged with DeepSeek V4 as described in Sec. A.4.

MathVision [25]. We evaluate the 304-example testmini split. The textual input is the dataset question itself, matching the public Qwen evaluation convention, without an additional answer-format instruction. Responses are judged by the same DeepSeek V4 correctness protocol used for MathVerse.

We-Math [17]. We evaluate the 1,740-example testmini split. The prompt presents the image, question, and answer options and requests an explicit reasoning trace followed by a clearly delimited final answer. We report accuracy from the benchmark's deterministic reasoning evaluator.

MathVista [14]. We evaluate the 1,000-example testmini split under three zero-shot prompting settings. The *CoT* setting asks for a reasoned answer followed by the required final value or option. The *Solution* setting explicitly prompts the model to produce a solution before the final answer. The *Format* setting requests only an answer in the format required by the target type, such as an option letter, integer, floating-point value with the prescribed precision, or list. Image captions and OCR annotations are not provided. We report the arithmetic mean over the three settings as the MathVista score.

MMK12-val [16]. MMK12 is an independently collected multimodal K–12 reasoning benchmark covering mathematics, physics, chemistry, and biology. We use its official 2,000-example test split as a fixed validation set during RL training. The same split and evaluation procedure are used for all compared methods within each model size. MMK12 does not participate in policy optimization, checkpoint selection, or early stopping. In Table 1, we report the score from the final scheduled validation event rather than selecting the best score observed during training.

A.2. Ablation configurations

Each variant modifies the indicated representation, metric, or allocation rule while retaining the remaining OVC procedure. Statistics are computed from the current actor and rollouts of the corresponding run, rather than copied from a separately trained full-OVC model. Representation and metric variants re-estimate their reference means, margin temperatures, refined references, and normalization factor. Within each update, the selected metric and pre-refinement temperatures remain fixed during the single reference refinement. All operations use valid response tokens and exclude padding.

Visual representations. Define the scaled affinities

$$a_{it}^{u} = \cos(S_i h_{it}, S_i g_i^{u}), \qquad u \in \{\mathrm{img}, \mathrm{ctx}\}. \tag{15}$$

Full OVC uses $\rho_{it} = a_{it}^{\mathrm{img}} - a_{it}^{\mathrm{ctx}}$. The representation variants replace the joint feature by

$$\begin{aligned} \text{No visual coordinate:} \quad & z_{it} = v_{it}, \\ \text{Absolute image affinity:} \quad & z_{it} = [v_{it}; a_{it}^{\mathrm{img}}], \\ \text{Separate affinity coordinates:} \quad & z_{it} = [v_{it}; a_{it}^{\mathrm{img}}; a_{it}^{\mathrm{ctx}}]. \end{aligned} \tag{16}$$

The separate-coordinate variant retains the same response-wise S_i for both affinities and uses an $(H+2)$-dimensional diagonal metric. Its shrinkage target remains the mean variation of the first H predictive coordinates; both appended affinity coordinates are excluded from that target but receive the same regularization in Eq. (10). "Absolute" means without context subtraction, not an absolute value. The $S_i = I$ variant removes scaling only from the two cosine measurements and retains their difference. In every representation variant, $v_{it} = (1 - p_{it})h_{it}$ is unchanged, and appended affinity coordinates are not multiplied by $1 - p_{it}$.

Table 3. Benchmark protocol used for external evaluation.

Benchmark	Split	Samples	Prompting protocol	Scoring protocol
MMMU	validation	900	multiple-choice prompting	deterministic accuracy
MathVerse	testmini	3,940	zero-shot direct query	DeepSeek V4 correctness judge
MathVision	testmini	304	original question	DeepSeek V4 correctness judge
We-Math	testmini	1,740	reasoning with explicit final answer	deterministic Loose accuracy
MathVista	testmini	1,000	three zero-shot prompting settings	answer extraction + deterministic scoring; mean over settings
MMK12-val	test	2,000	training validation prompt	validation accuracy

Metric variants. The identity variant sets $D = I$ over the entire selected feature space while retaining advantage-side reference estimation, soft refinement, and sign-conditioned allocation. The no-visual identity variant additionally sets $z_{it} = v_{it}$. For total-variance calibration, the side-normalized weights in Eq. (9) are retained, but both sides are centered at their common midpoint:

$$\begin{aligned}\overline{\mu}^{(0)} &= \frac{\mu_+^{(0)} + \mu_-^{(0)}}{2},\\ q_k^{\mathrm{tot}} &= \sum_{c\in\{-1,+1\}} \frac{1}{M_c} \sum_{(i,t)\in\mathcal{T}_c} |A_i| \big(z_{itk} - \overline{\mu}_k^{(0)}\big)^2 \\ &= q_k + \frac{1}{2}\big(\mu_{+,k}^{(0)} - \mu_{-,k}^{(0)}\big)^2.\end{aligned} \tag{17}$$

This is twice the total variance of an equally weighted mixture of the two advantage-side distributions, matching the summed-side scale of Eq. (9). It changes the centering without changing the relative side weights. The resulting q_k^{tot} and $\overline{q}^{\mathrm{tot}} = H^{-1}\sum_{k=1}^{H} q_k^{\mathrm{tot}}$ replace q_k and $\overline{q}$ in Eq. (10), with the same η, $r_{\min}$, and ϵ_D.

Allocation variants. At each actor update, these controls first compute the standard OVC normalized weights w_{it} for that run, then modify their assignment before evaluating Eq. (14). Response-mean allocation replaces every coefficient in response i by

$$w_{it}^{\mathrm{mean}} = \overline{w}_i = \frac{1}{T_i}\sum_{s=1}^{T_i} w_{is}. \tag{18}$$

Within-response permutation uses

$$w_{it}^{\mathrm{perm}} = w_{i,\Pi_i(t)}, \tag{19}$$

where Π_i is sampled uniformly over permutations of valid token positions, independently for each response at each actor update, and held fixed throughout that update.

Affinity-ranked allocation reassigns the same normalized weight multiset according to relative affinity ρ_{it}. Let $t_{i,1},\dots,t_{i,T_i}$ order the token positions by nondecreasing ρ_{it}, with ties resolved by token position, and let $w_{i,(1)} \le \cdots \le w_{i,(T_i)}$ be the sorted weights. Then

$$w_{i,t_{i,k}}^{\mathrm{aff}} = w_{i,(k)}, \qquad k = 1,\dots,T_i. \tag{20}$$

The ordering is independent of c_i: higher relative affinity receives a no-smaller coefficient on either advantage side. Only the assignment is changed; coefficient values remain those generated by OVC.

All three transformations preserve each response's total weight and hence the global token mean. Permutation and affinity ranking additionally preserve the complete within-response weight multiset. No second normalization is applied, and the selected coefficients remain detached in $A_{it}^{\prime,\mathrm{abl}} = \mathrm{sg}(w_{it}^{\mathrm{abl}})A_i$. Zero-advantage responses retain their constant normalized coefficients. These preservation properties hold within a fixed update, not across the evolving trajectories of separately trained variants (Appendix E.3).

A.3. Generation protocol

We use a common decoding configuration for all evaluations. Responses are sampled with temperature $T = 0.7$, nucleus-sampling threshold $p = 0.8$, and top-k truncation with $k = 20$. The presence penalty is 1.5 and the repetition penalty is 1.0. The default evaluation seed is 3407; additional replication seeds are reported with the corresponding results. The model's native thinking mode is disabled, although a benchmark prompt may still explicitly request a written reasoning trace.

The context window is 32,768 tokens, with at most 16,384 generated tokens per response. MathVision uses an image-area range of 602,112–4,014,080 pixels, and MMMU uses 1,003,520–4,014,080 pixels. The remaining benchmarks use the standard model processor settings. Benchmark-defined stop sequences are retained; otherwise generation terminates at the model's end-of-sequence token or the generation limit.

Benchmark-authored termination strings are retained, but the evaluator does not add an implicit double-newline stopping rule to tasks that do not define one. This is important for long-form mathematical responses, where stopping at the first blank line can truncate a valid multi-paragraph solution.

A.4. Scoring protocol

We use the DeepSeek V4 [4] model for evaluation where a language-model judge is required. The judging instructions and parsing rules are held fixed across all compared checkpoints.

MathVerse and MathVision. For each response, DeepSeek V4 receives the textual question, reference answer, and model response and returns a binary correctness decision. The judge assesses answer equivalence while allowing superficial differences in formatting, LaTeX notation, capitalization, and spacing. Numerical values, units, option identity, and required answer components must agree with the reference answer.

MathVista. DeepSeek V4 extracts the final answer from the model response using the question as context. The extracted answer is normalized and compared with the reference by the deterministic MathVista evaluator according to the target answer type, including multiple-choice and numerical answers. The reported MathVista score is the arithmetic mean over the three prompting settings.

MMMU and We-Math. MMMU is scored by its native deterministic accuracy evaluator. We-Math is scored using the benchmark's deterministic reasoning scorer.

A.5. Zero-shot VQA generalization

To examine whether the improvement of OVC extends beyond the benchmarks in our main evaluation, we additionally evaluate the final checkpoints on four zero-shot multimodal benchmarks: ChartQA [15], CharXiv [28], A-OKVQA [18], and SimpleVQA [3]. ChartQA and CharXiv primarily evaluate chart understanding and reasoning, whereas A-OKVQA and SimpleVQA evaluate more general visual question answering and multimodal factual knowledge.

For each model size, Base, GSPO, and OVC are evaluated under the same task-specific prompting, decoding, and scoring settings. No additional training or adaptation is performed on these four benchmarks. For CharXiv, we report the mean of its reasoning and descriptive scores. The Avg. bars in Figure 3 use the arithmetic mean of ChartQA, CharXiv, A-OKVQA, and SimpleVQA.

B. Relative Visual Features

B.1. Response-wise scaling and geometric interpretation

For coordinate k, the response-wise scale in Eq. (7) is

$$\bar{h}_{ik} = \frac{1}{T_i}\sum_t h_{itk}, \qquad \sigma_{ik} = \sqrt{\max\left\{\frac{1}{T_i}\sum_t (h_{itk}-\bar{h}_{ik})^2, \epsilon_\sigma^2\right\}} \tag{21}$$

Here $\epsilon_\sigma > 0$, and the variance is the population variance over valid response tokens. The same $S_i = \mathrm{diag}(\sigma_i)^{-1}$ transforms h_{it} and both prototypes, without subtracting $\bar{h}_i$ in the cosine. Only ρ_{it} uses this transform. The predictive feature remains $(1-p_{it})h_{it}$.

Let $W_i = S_i^\top S_i$. For nonzero transformed vectors and inactive numerical cosine floors,

$$\cos(S_i a, S_i b) = \frac{a^\top W_i b}{\sqrt{a^\top W_i a}\sqrt{b^\top W_i b}}. \tag{22}$$

Thus both affinities use the same response-dependent diagonal metric. Defining $\widehat{a} = S_i a/\|S_i a\|_2$ gives

$$\rho_{it} = \widehat{h}_{it}^\top\left(\widehat{g}_i^{\mathrm{img}} - \widehat{g}_i^{\mathrm{ctx}}\right), \qquad |\rho_{it}| \le 2. \tag{23}$$

The bound follows from Cauchy–Schwarz and the triangle inequality. Identical normalized prototypes yield zero relative affinity. Standard norm clamping also retains the bound since each clamped cosine remains in $[-1, 1]$.

This contrast does not remove all contextual information or isolate a causal image effect. Prompt states can already depend on the image. Image-position states follow causal visibility and need not encode question tokens appearing later in the prompt. The prototypes share a layer basis, not necessarily an identical information set. RVE provides one scalar per predictive state and does not localize image regions.

B.2. Predictive feature and gradient interpretation

Let the actual output-head input be f_t, with logits $a_j = W_j^\top f_t$ and probabilities $p_j = \mathrm{softmax}(a)_j$. Holding f_t fixed,

$$\log p_y = W_y^\top f_t - \log\sum_j \exp(W_j^\top f_t), \qquad \frac{\partial \log p_y}{\partial W_y} = (1-p_y)f_t. \tag{24}$$

This output-row identity motivates the uncertainty-scaled feature. In Eq. (8), replacing f_t with a state from the selected intermediate layer retains the factor $1-p_y$ but not the exact gradient identity. The final probabilities and intermediate states correspond to different stages of the actor. Accordingly, v_{it} is a gradient-inspired predictive feature, not a full-parameter score vector. The selected intermediate layer is fixed.

C. Properties of Outcome-conditioned Calibration

All results in this section condition on fixed reference means and a positive diagonal matrix D. They characterize the estimator at these statistics, rather than the evolution of the complete RL procedure. The reference estimators require $M_+ > 0$ and $M_- > 0$. Under $0 \le \eta \le 1$, $r_{\min} \ge 0$, and $\epsilon_D > 0$, Eq. (10) ensures $d_k \ge \epsilon_D > 0$ because $q_k \ge 0$, and hence guarantees $D \succ 0$. A missing advantage side is outside the domain of this two-reference estimator; a denominator floor alone does not define the missing empirical reference.

Table 4. **Per-task zero-shot transfer results.** CharXiv is the mean of its reasoning and descriptive scores. The corresponding gains over Base are visualized in Figure 3.

Method	ChartQA	CharXiv	A-OKVQA	SimpleVQA	Avg.
Qwen3-VL-4B					
Base	83.84	58.04	84.89	25.93	63.18
GSPO [42]	84.12	59.68	85.41	25.98	63.80
GSPO+OVC (Ours)	84.16	60.12	86.03	25.43	63.94
Qwen3-VL-8B					
Base	85.24	65.68	87.95	27.70	66.64
GSPO [42]	84.68	66.44	88.30	27.01	66.61
GSPO+OVC (Ours)	85.65	66.98	88.61	28.06	67.33
Qwen3.5-9B					
Base	85.36	75.09	87.25	27.75	68.86
GSPO [42]	85.56	74.96	87.69	27.75	68.99
GSPO+OVC (Ours)	86.43	75.54	87.42	27.80	69.30

C.1. Optimal coefficients and normalized contrast

Proposition 1 (Regularized contrast)**.** *Let $D \succ 0$ and $\Delta\mu = \mu_+ - \mu_-$. The unique maximizer of*

$$F_D(b) = b^\top \Delta\mu - \frac{1}{2} b^\top D b \tag{25}$$

is $b^* = D^{-1}\Delta\mu$. If $\Delta\mu \neq 0$, the nonzero scalar multiples of b^* are exactly the maximizers of

$$\mathcal{R}_D(b) = \frac{(b^\top \Delta\mu)^2}{b^\top D b}, \qquad b \neq 0, \qquad \max_b \mathcal{R}_D(b) = \Delta\mu^\top D^{-1}\Delta\mu. \tag{26}$$

Proof. Differentiating the quadratic criterion gives

$$\nabla_b F_D(b) = \Delta\mu - Db, \qquad \nabla_b^2 F_D(b) = -D \prec 0. \tag{27}$$

Thus F_D is strictly concave, and its unique stationary point $b^* = D^{-1}\Delta\mu$ is the unique maximizer. For the normalized criterion, write $u = D^{1/2}b$ and $a = D^{-1/2}\Delta\mu$. Cauchy–Schwarz gives

$$\mathcal{R}_D(b) = \frac{(u^\top a)^2}{u^\top u} \leq a^\top a = \Delta\mu^\top D^{-1}\Delta\mu. \tag{28}$$

Equality holds when b is a nonzero scalar multiple of $D^{-1}\Delta\mu$. □

When $\Delta\mu = 0$, F_D is uniquely maximized at $b = 0$, whereas $\mathcal{R}_D(b) = 0$ for every nonzero b. The denominator in $\mathcal{R}_D$ is a regularized diagonal variation surrogate, not generally the full empirical variance of $b^\top z$, since cross-coordinate covariances are not modeled. The result requires neither a Gaussian assumption nor a fitted probability model.

Comparison with an uncalibrated direction. For $b_0 = \Delta\mu \neq 0$, Proposition 1 yields

$$\mathcal{R}_D(b^*) = \Delta\mu^\top D^{-1}\Delta\mu \geq \frac{\|\Delta\mu\|_2^4}{\Delta\mu^\top D \Delta\mu} = \mathcal{R}_D(b_0). \tag{29}$$

Equality holds exactly when $\Delta\mu$ is an eigenvector of D. This comparison characterizes coordinate calibration under the stated fixed-statistic criterion. It does not establish a corresponding ordering of rewards, token-label accuracy, or generalization after reference refinement and nonlinear weight mapping.

C.2. Equivalence to a distance contrast

With $\overline{\mu} = (\mu_+ + \mu_-)/2$ and $\|a\|_{D^{-1}}^2 = a^\top D^{-1} a$,

$$\begin{aligned} &\|z - \mu_-\|_{D^{-1}}^2 - \|z - \mu_+\|_{D^{-1}}^2 \\ &= 2z^\top D^{-1}(\mu_+ - \mu_-) + \mu_-^\top D^{-1}\mu_- - \mu_+^\top D^{-1}\mu_+ \\ &= 2\Delta\mu^\top D^{-1}(z - \overline{\mu}) = e(z). \end{aligned} \tag{30}$$

This identity follows directly from cancellation of the common quadratic term and does not require the variational characterization in Proposition 1. A large coordinate magnitude alone therefore does not determine the distance contrast. The calibration rescales a nonzero outcome difference by d_k^{-1}, while coordinates with equal reference means contribute zero directly. With $D = I$, this score becomes an ordinary squared-distance contrast.

C.3. Direct contribution of the visual coordinate

For $z = [v; \rho]$ and $D = \mathrm{diag}(D_h, d_\rho)$, Eq. (30) yields

$$e(z) = 2\Delta\mu_h^\top D_h^{-1}(v - \overline{\mu}_h) + 2\frac{\Delta\mu_\rho}{d_\rho}(\rho - \overline{\mu}_\rho), \tag{31}$$

which proves Eq. (5). The maximum criterion in Eq. (26) also decomposes as

$$\mathcal{R}_D^{\max} = \Delta\mu_h^\top D_h^{-1}\Delta\mu_h + \frac{(\Delta\mu_\rho)^2}{d_\rho}. \tag{32}$$

The last term quantifies the visual coordinate's contribution to this fixed-statistic diagonal criterion. It is not a measure of information unique to the image, since redundancy with hidden coordinates is not removed. Removing RVE and refitting the estimator can change the hidden references as well.

Holding the final references, D, $\tau_c > 0$, the predictive feature v, and the advantage side $c \in \{-1, +1\}$ fixed, let $u_c = \mathrm{sigmoid}(ce^*(z)/\tau_c)$. Then

$$\frac{\partial e^*}{\partial \rho} = \frac{2\Delta\mu_\rho^*}{d_\rho}, \qquad \frac{\partial u_c}{\partial \rho} = u_c(1-u_c)\frac{2c\Delta\mu_\rho^*}{\tau_c d_\rho}. \tag{33}$$

The raw weight has the same sensitivity multiplied by $\lambda_{\max} - \lambda_{\min}$. These derivatives show that the response's advantage side and the empirical visual contrast jointly determine the dependence on visual affinity. These are conditional sensitivities of the evidence and raw coefficient map, not gradients propagated through the credit estimator. The final normalized coefficients additionally depend on the shared factor Z, and re-estimating the reference statistics introduces further dependencies. Here calibration refers to variance scaling of observed advantage-side contrasts, not correctness-probability calibration or causal attribution.

D. Soft Reference Refinement

The default estimator performs one refinement. Let $m_{it}^{(0)} = c_i e^{(0)}(z_{it})$, and compute

$$\overline{m}_c = \frac{1}{|\mathcal{T}_c|}\sum_{\mathcal{T}_c} m_{it}^{(0)}, \qquad \tau_c = \max\left\{\sqrt{\frac{1}{|\mathcal{T}_c|}\sum_{\mathcal{T}_c}(m_{it}^{(0)} - \overline{m}_c)^2}, \epsilon_\tau\right\}. \tag{34}$$

Here $\epsilon_\tau > 0$. Unlike the reference moments, this population standard deviation is not weighted by $|A_i|$. The temperatures are computed from the initial references before Eq. (11) updates those references once. Final weights use the refined references while retaining both the initial D and the pre-refinement τ_c.

Entropy-regularized assignment. The soft assignment has the following standard variational form [40]. For fixed m and $\tau > 0$, consider

$$\max_{u\in[0,1]} \{um + \tau\mathcal{H}(u)\}, \qquad \mathcal{H}(u) = -u\log u - (1-u)\log(1-u). \tag{35}$$

With the convention $0\log 0 = 0$, the objective is continuous on $[0, 1]$. On $(0, 1)$, its first derivative is $m + \tau\log((1-u)/u)$ and its second derivative is $-\tau/[u(1-u)] < 0$. For finite m, the unique stationary point lies in $(0, 1)$ and is the unique maximizer:

$$u = \mathrm{sigmoid}(m/\tau). \tag{36}$$

Reference update. Fix $r_{it}^c = |A_i|u_{it}^{(0)}$ for $(i, t) \in \mathcal{T}_c$. For finite margins, $0 < u_{it}^{(0)} < 1$, so $M_c > 0$ implies $\sum_{\mathcal{T}_c} r_{it}^c > 0$ in exact arithmetic. Consider the weighted least-squares problem

$$\min_\mu \sum_{\mathcal{T}_c} r_{it}^c \|z_{it} - \mu\|_{D^{-1}}^2. \tag{37}$$

Its Hessian is $2(\sum_{\mathcal{T}_c} r_{it}^c)D^{-1} \succ 0$. Setting its gradient to zero therefore gives the unique minimizer:

$$2D^{-1}\sum_{\mathcal{T}_c} r_{it}^c(\mu - z_{it}) = 0, \qquad \mu = \frac{\sum_{\mathcal{T}_c} r_{it}^c z_{it}}{\sum_{\mathcal{T}_c} r_{it}^c}, \tag{38}$$

recovering Eq. (11). These variational characterizations apply separately to the assignment and reference-update subproblems. They do not identify the complete procedure as EM or as coordinate ascent on a common objective. The estimator performs one refinement rather than iterating to convergence.

E. Credit Coefficients and Policy Updates

E.1. Normalization, bounds, and signs

Assume $N > 0$ and $0 < \lambda_{\min} \le \lambda_{\max}$. Every raw coefficient, including that of a zero-advantage token, lies in $[\lambda_{\min}, \lambda_{\max}]$. Its global token mean Z therefore lies in the same positive interval. Consequently,

$$\frac{1}{N}\sum_{i,t} w_{it} = 1, \qquad \frac{\lambda_{\min}}{\lambda_{\max}} \le w_{it} \le \frac{\lambda_{\max}}{\lambda_{\min}}, \qquad \mathrm{sign}(w_{it}A_i) = \mathrm{sign}(A_i). \tag{39}$$

The raw interval $[0.8, 1.2]$ thus gives the conservative final bounds $[2/3, 3/2]$. Zero-advantage tokens affect Z but make no direct policy-gradient contribution. Global token-mean normalization does not preserve each response's mean coefficient, the advantage mass of either side, or the policy-gradient norm.

E.2. Token-specific gradients under sequence clipping

The stop-gradient construction in Eq. (13) gives

$$\widetilde{s}_{it} = s_i \quad \text{numerically}, \qquad \nabla_\theta \widetilde{s}_{it} = s_i \nabla_\theta L_{it}(\theta). \tag{40}$$

This is the GSPO-token derivative. For nonzero advantages and away from clipping boundaries, define

$$\kappa_i = \begin{cases} \mathbb{I}\{s_i < 1 + \epsilon_h\}, & A_i > 0, \\ \mathbb{I}\{s_i > 1 - \epsilon_l\}, & A_i < 0. \end{cases} \tag{41}$$

All valid tokens in a response have the same numerical ratio and advantage sign. The active clipping branch is therefore shared across that response. Treating A_i and w_{it} as constants during differentiation,

$$\nabla_\theta \mathcal{J}_{\mathrm{OVC}} = \frac{1}{B} \sum_{i \mid A_i \neq 0} \frac{\kappa_i s_i A_i}{T_i} \sum_t w_{it} \nabla_\theta L_{it}(\theta). \tag{42}$$

Sequence clipping therefore selects the active response-level surrogate terms, while each retained token-gradient term is multiplied by its own coefficient. Setting all $w_{it} = 1$ recovers the gradient of the corresponding GSPO surrogate with the same response averaging and clipping rule.

Why a shared differentiable ratio is insufficient. If one directly reuses differentiable s_i in every token term, the unclipped objective becomes

$$\mathcal{J}_{\mathrm{shared}} = \frac{1}{B} \sum_i A_i s_i \overline{w}_i, \qquad \overline{w}_i = \frac{1}{T_i} \sum_t w_{it}. \tag{43}$$

Its gradient is

$$\nabla_\theta \mathcal{J}_{\mathrm{shared}} = \frac{1}{B} \sum_i \frac{s_i A_i \overline{w}_i}{T_i} \sum_t \nabla_\theta L_{it}(\theta). \tag{44}$$

Only the response-average coefficient survives. The same reduction holds under clipping, with the corresponding factor κ_i. The stop-gradient construction instead preserves the correspondence between individual tokens and their weights, despite an identical forward ratio value.

E.3. Separating response weighting from token allocation

For fixed rollouts and a fixed actor parameter value θ, let $g_{it} = \nabla_\theta L_{it}(\theta)$ and $\overline{g}_i = T_i^{-1} \sum_t g_{it}$. Define the vector

$$C_i = \frac{1}{T_i} \sum_t (w_{it} - \overline{w}_i)(g_{it} - \overline{g}_i). \tag{45}$$

Expanding and using the zero sums of centered quantities yields

$$\frac{1}{T_i} \sum_t w_{it} g_{it} = \overline{w}_i \, \overline{g}_i + C_i, \qquad \nabla_\theta \mathcal{J}_{\mathrm{OVC}} = \frac{1}{B} \sum_{i \mid A_i \neq 0} \kappa_i s_i A_i (\overline{w}_i \overline{g}_i + C_i) \tag{46}$$

The first term is response-level weighting. The second depends on the correspondence between weights and token gradients within that response. Replacing every weight by $\overline{w}_i$ removes C_i exactly while preserving the first term.

For a uniformly random permutation Π_i of a response's weights, with token gradients and the weight multiset held fixed,

$$\mathbb{E}_{\Pi_i}[C_i^{\Pi_i}] = 0, \tag{47}$$

because $\mathbb{E}_{\Pi_i}[w_{i,\Pi_i(t)}] = \overline{w}_i$ at every position. This conditional expectation concerns a fixed update; an individual permutation need not eliminate the allocation term. It motivates permutation and response-mean controls without implying identical training trajectories or long-run performance.

The preceding identities characterize the detached surrogate in Eq. (14). They do not make the coefficients ground-truth process advantages, establish an unbiased gradient of the original expected reward, or guarantee monotonic improvement. Shared model parameters also prevent advantage-sign preservation from implying an independent probability increase or decrease for every token.

F. Algorithm and Implementation

F.1. Complete algorithm and implementation details

Statistical scope and alignment. The described recipe uses one image per input, nonempty image and context position sets, and a fixed intermediate layer. The state h_{it} is taken at the position predicting y_{it}, rather than at the position after y_{it} has been consumed. Probabilities come from the current actor rather than being reused from the behavior policy. In the default micro-batch of one response, the RVE scale is response-wise even though $G = 8$.

Reference moments, margin scales, and normalization are aggregated over the actor mini-batch across micro-batches and data-parallel ranks. Initial sufficient statistics consist of each side's mass, weighted feature sum, and weighted squared-feature sum. Margin temperatures use unweighted token counts and margin moments. RVE scales are computed before these aggregations and are not global rollout-group statistics.

Nonzero and degenerate batches. The mathematical reference formulas assume positive side masses. For complete rollout groups, any group with nonconstant rewards supplies both advantage signs. If all groups have constant rewards, every A_i and the policy gradient are zero. Numerical denominators and variances use floors in the implementation. Group reward standard deviations use the sample estimator with denominator $G-1$, whereas the feature and margin variances use population estimators. The all-image formulation does not describe mixed image-free data.

Normalization domains. The reference estimator uses $|A_i|$ for each token rather than $|A_i|/T_i$. Longer responses therefore have greater total weight in the reference statistics. In contrast, Eq. (14) averages tokens within a response and then averages responses. Global coefficient normalization does not remove this difference.

Algorithm 1 OVC actor update with one reference refinement

Require: Rollouts, masks, rollout-group rewards, behavior log probabilities, and actor θ.
Require: $N > 0$ and $M_+, M_- > 0$.
Compute A_i and c_i using Eq. (1).
Keep θ fixed until the final optimizer step.
All aggregates span valid actor-mini-batch tokens across micro-batches and data-parallel ranks.
Passes 1–4 recompute z_{it} using response-wise RVE and Eq. (8), without gradients.
Pass 1: Reference moments
Aggregate, for $c \in \{-1, +1\}$ and $k = 1, \ldots, H+1$, $(M_c, S_{c,k}, Q_{c,k}) \leftarrow \sum_{(i,t)\in\mathcal{T}_c} |A_i|(1, z_{itk}, z_{itk}^2)$.
$\mu_{c,k}^{(0)} \leftarrow S_{c,k}/M_c$, $q_k \leftarrow \sum_{c\in\{-1,+1\}} [Q_{c,k}/M_c - (\mu_{c,k}^{(0)})^2]$.
Construct D using Eq. (10).
Pass 2: Margin scales
Compute $c_i e^{(0)}(z_{it})$ using $\mu_{\pm}^{(0)}$ and D.
Estimate τ_c from unweighted margin moments using Eq. (34).
Pass 3: Reference refinement
Compute $u_{it}^{(0)}$ and aggregate μ_c^* using Eq. (11).
Form e^* with $\mu_{\pm}^*$; retain D and the pre-refinement τ_c.
Pass 4: Weight normalization
Compute $\widehat{w}_{it}$ using Eq. (12); set $\widehat{w}_{it} = \lambda_{\min}$ when $A_i = 0$.
$Z \leftarrow N^{-1} \sum_{i,t} \widehat{w}_{it}$, including zero-advantage tokens.
Pass 5: Policy optimization
Recompute differentiable $L_{it}(\theta)$ and detached credit features.
Recompute $w_{it} = \widehat{w}_{it}/Z$ with $\mu_{\pm}^*, D, \tau_c, Z$ fixed.
Set $A'_{it} = \mathrm{sg}(w_{it})A_i$ and construct GSPO-token ratios using Eq. (13).
Accumulate gradients of $-\mathcal{J}_{\mathrm{OVC}}$ using Eq. (14); update θ once.

Computation. For one refinement, the current implementation uses four no-gradient statistics forwards and one training forward per micro-batch. The actor parameters do not change between these passes. Sufficient-statistic aggregation agrees algebraically with direct global aggregation when the recomputed features are identical, up to floating-point order. The calibration arithmetic is $O(NH)$ per pass with $O(H)$ accumulated statistics, excluding model forwards, activation storage, and communication. These counts are not measured runtime or speedup claims. Standard inference does not require the credit estimator.

The supplied reward combines accuracy and format terms as $R_i = 0.9R_i^{\mathrm{acc}} + 0.1R_i^{\mathrm{fmt}}$. Consequently, advantage signs should not be described as pure correctness labels. The stated GSPO objective has no dual-clipping term. Neither adaptive layer selection, region supervision, a full covariance estimator, nor an extra process reward model is part of this method.

F.2. Training recipes and checkpoints

Table 5 summarizes the training recipes and checkpoints used for the main comparisons. Within each model setting, the compared methods use the same training data and rollout configuration unless otherwise noted. The reported main results use the fixed checkpoints listed in the table.

Additional OVC-specific implementation constants for the Qwen3.5-9B recipe are listed in Table 6. These settings concern the credit estimator itself and are separate from the model-level training recipe summarized above.

Table 5. Training recipes and evaluated checkpoints for the three model settings.

Model	Training data	Prompt batch	Rollouts G	Learning rate	Max prompt	Max response	Checkpoint
Qwen3-VL-4B	ViRL39K	512	16	1×10^{-6}	2,048	20,480	step 76
Qwen3-VL-8B	ViRL39K	512	16	1×10^{-6}	2,048	8,192	step 76
Qwen3.5-9B	Math77K	256	8	1×10^{-6}	4,096	16,384	step 300

Table 6. Defaults of the supplied Qwen3.5-9B recipe.

Quantity	Setting
Predictive layer and hidden width	$\ell = 16$, $H = 4096$
Responses per input	$G = 8$
Global actor mini-batch	512 responses
Forward micro-batch per GPU	1 response
Reference refinements	$K = 1$
Raw coefficient interval	$[0.8, 1.2]$
Shrinkage and relative variance floor	$\eta = 0.1$, $r_{\min} = 10^{-3}$
Response scale floor	$\epsilon_\sigma = 10^{-6}$
Margin temperature floor	$\epsilon_\tau = 10^{-8}$
Lower and upper ratio clipping	3×10^{-4}, 4×10^{-4}
Auxiliary KL and entropy losses	Disabled